Freight Rate Prediction

Machine Learning Project Report

Prepared as a concise summary of the modeling workflow and evaluation.

1. Overview

This project builds a regression model to estimate freight rates from shipment and market-related information. CatBoost Regressor was selected because it handles categorical features such as pickup, delivery, and equipment type without requiring manual one-hot encoding.

2. Dataset

The labeled training dataset contains 48,000 records with the target column posted_rate. A separate validation dataset contains 12,000 records without target values. The December chart input contains 31 rows for December 1–31, 2025.

Available inputs include pickup and delivery locations, coordinates, distance, equipment, weight, date, market index, and quote signal. The December input contains a smaller set of shared shipment features.

3. Data Preparation

The load identifier is excluded from the model features. The date column is converted into month, day, day of week, and day of year features. Negative weight values are treated as missing values, while missing categorical values are filled with 'Unknown'.

The labeled data is split chronologically: records before October 1, 2025 are used for training, and October 2025 records are used for validation. This gives a time-based check rather than a random split.

4. Model and Evaluation

A CatBoostRegressor model is trained on the log-transformed target using log1p. Predictions are converted back to the original rate scale with expm1. This approach was tested because freight rates include a small number of unusually high values.

On the October holdout, the log-target model achieved MAE 108.21, RMSE 647.71, and R^2 0.8204. The baseline model achieved MAE 118.35, RMSE 647.61, and R^2 0.8205. The log-target model reduced MAE, while RMSE and R^2 remained very similar.

5. Prediction Outputs

The validation prediction file contains 12,000 rows with load_id and predicted_rate. A separate December model is trained using only the features available in the December input file, then generates one predicted rate for each of the 31 dates.

The prediction outputs are checked for expected row counts, matching validation IDs, finite numeric values, and positive predicted rates.

6. Limitations

The target distribution contains rare high-rate cases, which contribute strongly to RMSE. The October holdout is a useful time-based validation, but performance may differ on other periods or routes.

The December model uses fewer features than the main validation model because the December input does not include every training column. Its predictions should therefore be interpreted separately from the October validation metrics.

7. Conclusion

The project provides an end-to-end baseline for freight-rate estimation: data checks, feature engineering, chronological validation, CatBoost training, feature importance, and prediction file generation. Further improvements could include investigating high-rate outliers, evaluating additional time splits, and checking performance by equipment type and route.

Key October holdout metrics

Model	MAE	RMSE	R^2
CatBoost baseline	118.35	647.61	0.8205
CatBoost with log target	108.21	647.71	0.8204